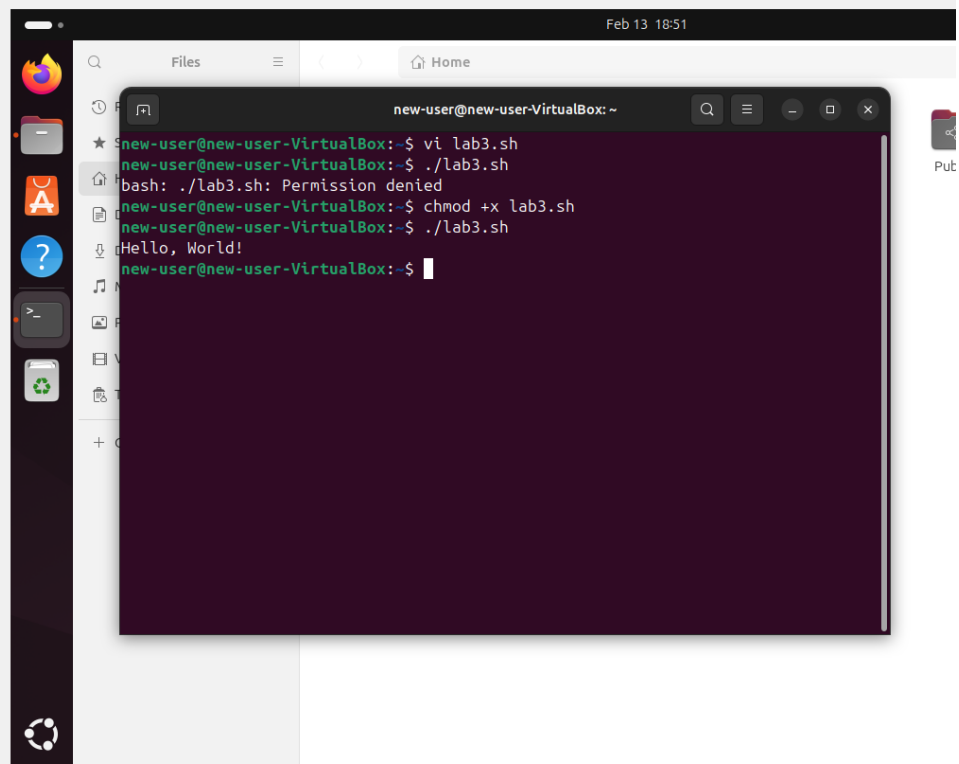


This tells that first we created through script .sh file then press I then wrote #bin etc then text and then esc and then :wq hen enter then printed hello world after getting permisiion.



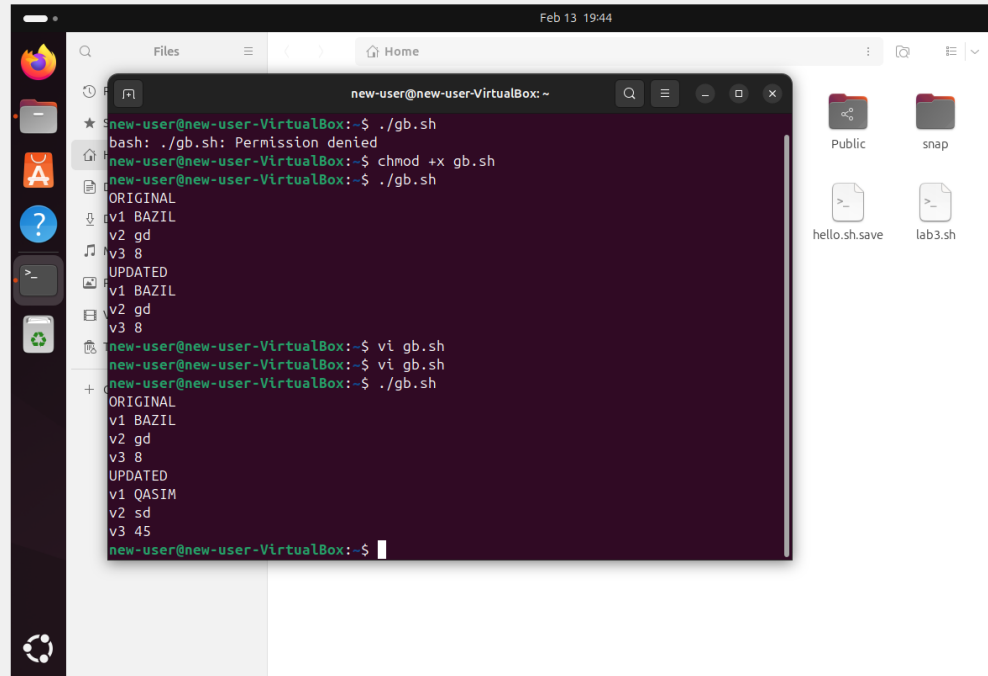
The screenshot shows a terminal window titled "new-user@new-user-VirtualBox: ~" with a dark purple background. The terminal displays the following commands and output:

```
new-user@new-user-VirtualBox:~$ vi lab3.sh
new-user@new-user-VirtualBox:~$ ./lab3.sh
bash: ./lab3.sh: Permission denied
new-user@new-user-VirtualBox:~$ chmod +x lab3.sh
new-user@new-user-VirtualBox:~$ ./lab3.sh
Hello, World!
new-user@new-user-VirtualBox:~$
```

The terminal window is overlaid on a desktop environment. In the background, a "Files" window is visible with a sidebar containing icons for Home, Recent, and other locations. The desktop also features a dock with various application icons on the left and a "Put" button on the right. The top of the desktop shows a date and time indicator: "Feb 13 18:51".

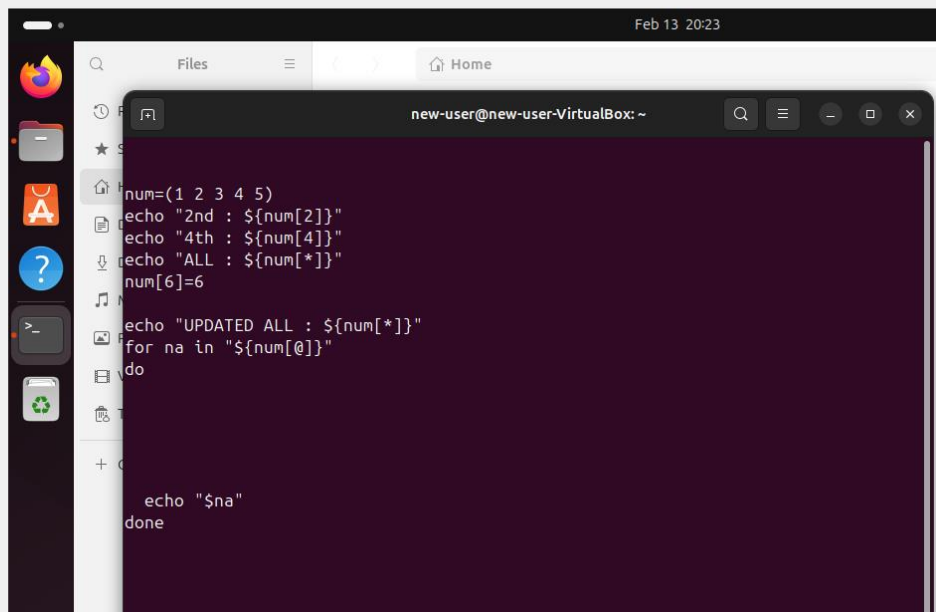
TASK#2).

Ubuntu 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help



```
new-user@new-user-VirtualBox: ~  
$ ./gb.sh  
bash: ./gb.sh: Permission denied  
new-user@new-user-VirtualBox:~$ chmod +x gb.sh  
new-user@new-user-VirtualBox:~$ ./gb.sh  
ORIGINAL  
v1 BAZIL  
v2 gd  
v3 8  
UPDATED  
v1 BAZIL  
v2 gd  
v3 8  
new-user@new-user-VirtualBox:~$ vi gb.sh  
new-user@new-user-VirtualBox:~$ vi gb.sh  
new-user@new-user-VirtualBox:~$ ./gb.sh  
ORIGINAL  
v1 BAZIL  
v2 gd  
v3 8  
UPDATED  
v1 QASIM  
v2 sd  
v3 45  
new-user@new-user-VirtualBox:~$
```

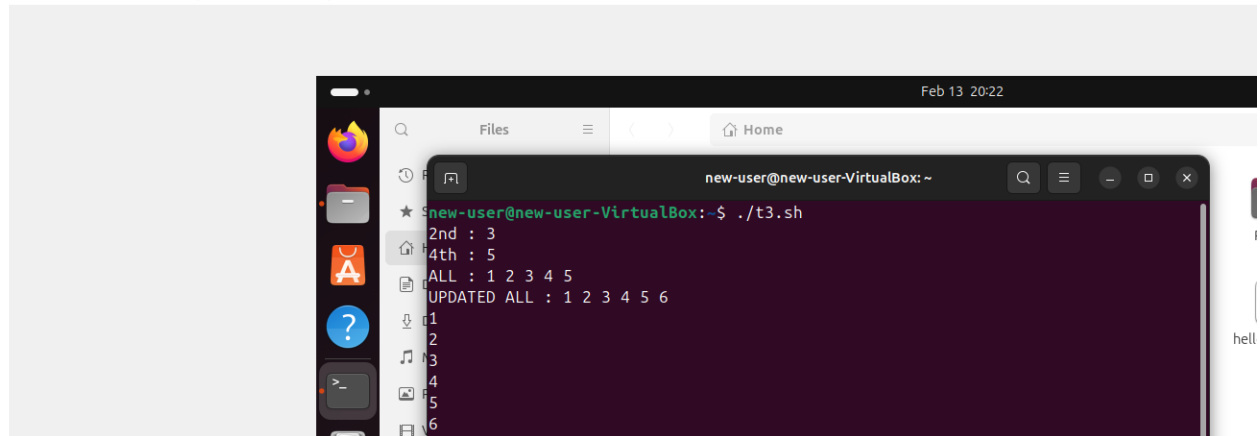
Ubuntu 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help



```
new-user@new-user-VirtualBox: ~  
$ num=(1 2 3 4 5)  
$ echo "2nd : ${num[2]}"  
$ echo "4th : ${num[4]}"  
$ echo "ALL : ${num[*]}"  
$ num[6]=6  
$ echo "UPDATED ALL : ${num[*]}"  
$ for na in "${num[@]}"  
$ do  
$  
$ echo "$na"  
$ done
```

Ubuntu 1 [Running] - Oracle VirtualBox

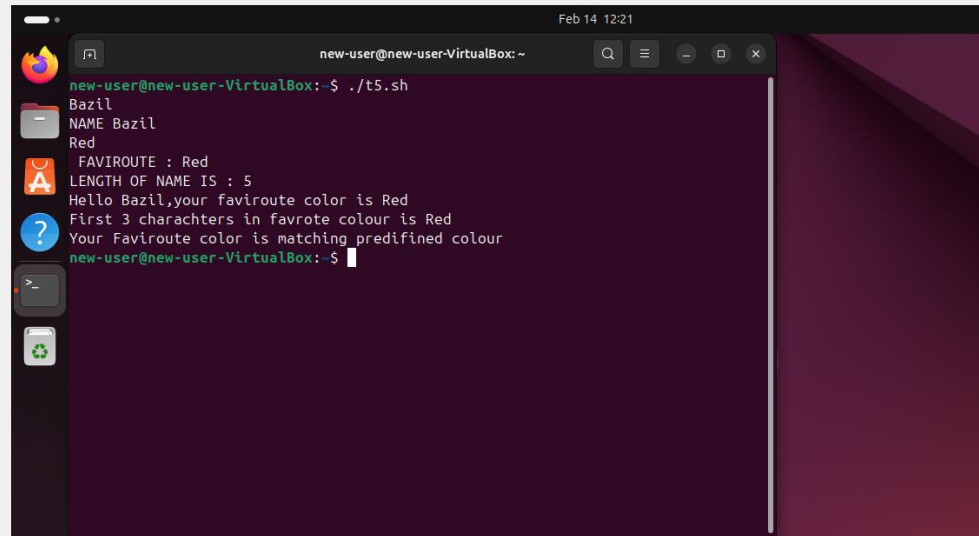
File Machine View Input Devices Help



TASK#3).

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help



```
new-user@new-user-VirtualBox: ~  
new-user@new-user-VirtualBox:~$ ./t5.sh  
Bazil  
NAME Bazil  
Red  
FAVIROUTE : Red  
LENGTH OF NAME IS : 5  
Hello Bazil,your faviroute color is Red  
First 3 charachters in favrote colour is Red  
Your Faviroute color is matching predefined colour  
new-user@new-user-VirtualBox:~$
```

```

new-user@new-user-VirtualBox: ~
Feb 14 12:22

read name
echo "NAME $name "

read favioo

echo " FAVIROUTE : $favioo"
echo "LENGTH OF NAME IS : ${#name}"

echo "Hello $name,your faviroute color is $favioo"
echo "First 3 charachters in favrote colour is ${favioo:0:3}"

definedal="Red"

```

```

new-user@new-user-VirtualBox: ~
Feb 14 12:22

read favioo

echo " FAVIROUTE : $favioo"
echo "LENGTH OF NAME IS : ${#name}"

echo "Hello $name,your faviroute color is $favioo"
echo "First 3 charachters in favrote colour is ${favioo:0:3}"

definedal="Red"

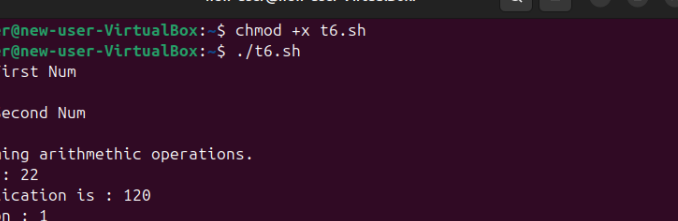
if [ "$favioo" == "$definedal" ]
then
echo "Your Faviroute color is matching predifined colour "
else
echo "Your faviroute colour didnt match the predefined colour."
fi

```

TASK#4).

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help



The screenshot shows a terminal window titled "new-user@new-user-VirtualBox: ~". The window has a dark theme with a purple background. On the left side, there is a vertical dock with icons for Firefox, a file manager, an application store, a help icon, a terminal, and a system monitor. The terminal text is as follows:

```
new-user@new-user-VirtualBox:~$ chmod +x t6.sh
new-user@new-user-VirtualBox:~$ ./t6.sh
Enter First Num
12
Enter Second Num
10
Performing arithmetic operations.
Sum is : 22
Multiplication is : 120
Division : 1
Mod is : 2
First number is greater than second
Subtraction is : 2
new-user@new-user-VirtualBox:~$
```

```

new-user@new-user-VirtualBox: ~
#!/bin/bash
echo "Enter First Num"
read num1
echo "Enter Second Num"
read num2
echo "Performing arithmetic operations."

echo "Sum is : $((num1+num2))"
echo "Multiplication is : $((num1*num2))"
"t6.sh" 42 lines, 588 bytes

```

```

new-user@new-user-VirtualBox: ~
echo "Sum is : $((num1+num2))"
echo "Multiplication is : $((num1*num2))"
if [ "$num2" -ne 0 ]
then
    echo "Division : $((num1 / num2))"
    echo "Mod is : $((num1%num2))"
else
    echo "Division not possible"
fi
if [ "$num1" == "$num2" ]
then
    echo "Both Numbers are Equal"
elif [ "$num1" -gt "$num2" ]
then
    echo "First number is greater than second "
else
    echo "First number is less than second "
fi
echo "Subtraction is : $((num1-num2))"

```


TASK#5).

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
new-user@new-user-VirtualBox: ~  
new-user@new-user-VirtualBox:~$ vi t7.sh  
new-user@new-user-VirtualBox:~$ vi t7.sh  
new-user@new-user-VirtualBox:~$ chmod +x t7.sh  
new-user@new-user-VirtualBox:~$ ./t7.sh  
Menu Driven Sytem.  
1: Display Date.  
2: Listing Files.  
3: Show current directory.  
4: exit.  
7  
Invalidated User Input.  
new-user@new-user-VirtualBox:~$ ./t7.sh  
Menu Driven Sytem.  
1: Display Date.  
2: Listing Files.  
3: Show current directory.  
4: exit.  
3  
Validated.  
/home/new-user  
new-user@new-user-VirtualBox:~$ ./t7.sh  
Menu Driven Sytem.  
1: Display Date.  
2: Listing Files.
```

Ubuntu 1 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
new-user@new-user-VirtualBox: ~  
7  
Invalidated User Input.  
new-user@new-user-VirtualBox:~$ ./t7.sh  
Menu Driven Sytem.  
1: Display Date.  
2: Listing Files.  
3: Show current directory.  
4: exit.  
3  
Validated.  
/home/new-user  
new-user@new-user-VirtualBox:~$ ./t7.sh  
Menu Driven Sytem.  
1: Display Date.  
2: Listing Files.  
3: Show current directory.  
4: exit.  
2  
Validated.  
Desktop gb.sh h.sh Pictures t3.sh t6.sh  
Documents gh1.sh lab3.sh Public t3..sh t7.sh  
Downloads gh.sh Music snap t4.sh Templates  
g2.sh hello.sh.save os_lab_24k_0559 t2.sh t5.sh Videos  
new-user@new-user-VirtualBox:~$
```

```
new-user@new-user-VirtualBox: ~  
#!/bin/bash  
echo "Menu Driven Sytem."  
echo "1: Display Date."  
echo "2: Listing Files."  
echo "3: Show current directory."  
echo "4: exit."  
read choice  
if [ "$choice" -gt 0 ] && [ "$choice" -lt 5 ]  
then  
echo "Validated. "  
  
case "$choice" in  
1) date;;  
2) ls;;  
3) pwd;;  
"t7.sh" 37 lines, 386 bytes
```

```
new-user@new-user-VirtualBox: ~  
echo "Validated. "  
  
case "$choice" in  
1) date;;  
2) ls;;  
3) pwd;;  
4) echo "Error"  
exit 0;;  
esac  
  
else  
echo "Invalidated User Input."  
fi
```